

Because $\Delta\psi_g$ is the change in bearing that occurs along a path of zero acceleration, this change in heading does not result from rotation of the aircraft. It results simply from a change in direction of the local meridian relative to the local geodesic, which occurs whenever an aircraft travels any path with an east-west component that does not follow exactly along the equator. Thus, **the geodesic change in bearing that is obtained from Eq. (11.12.37) is added to any change in heading that is predicted from integration of the aircraft equations of motion in local flat-Earth coordinates.**

Because the Euler-Rodrigues quaternion is used to describe aircraft orientation in Eqs. (11.11.1) through (11.11.4), we must be able to evaluate the change in the quaternion that results from a geodesic change in bearing. Using the algorithms that are presented in Sec. 11.7, this could be done by transforming the flat-Earth quaternion to Euler angles, adding the geodesic change in bearing to the azimuth angle, and then transforming the result back to a quaternion. However, this would be very inefficient. A more efficient algorithm is acquired by rearranging results obtained from Eqs. (11.6.3), (11.6.13), and (11.7.8). This yields the **quaternion transformation from flat-Earth coordinates to geographic coordinates**

$$\begin{Bmatrix} e_0 \\ e_x \\ e_y \\ e_z \end{Bmatrix}_{\text{geographic}} = \cos(\Delta\psi_g/2) \begin{Bmatrix} e_0 \\ e_x \\ e_y \\ e_z \end{Bmatrix}_{\text{flat Earth}} + \sin(\Delta\psi_g/2) \begin{Bmatrix} -e_z \\ -e_y \\ e_x \\ e_0 \end{Bmatrix}_{\text{flat Earth}} \quad (11.12.38)$$

where $\Delta\psi_g$ is the geodesic bearing change, which could be obtained from the small-angle ellipsoidal-Earth approximation given by Eq. (11.12.21) or from the analytical solution for the spherical-Earth model that is expressed in Eq. (11.12.37). The development of Eq. (11.12.38) is left as an exercise for the student.

Within the constant-curvature spherical-Earth model, **Eqs. (11.12.37) and (11.12.38) are mathematically exact and singularity free. These relations assume only that the change in altitude over the flight segment is small compared with the radius of the Earth. They do not assume small changes in latitude and longitude.** Thus, the relations given by Eqs. (11.12.37) and (11.12.38) provide a closed-form solution for level flight. Nevertheless, numerical computations for small changes in aircraft position in geographic coordinates require evaluating small differences in large numbers. The Earth's radius is on the order of 10^8 ft. Double-precision computations provide results with 14 to 15 significant digits. Thus, double-precision computations in geographic coordinates result in a net precision on the order of 10^{-6} ft. This is approximately equivalent to the precision obtained from single-precision computations in flat-Earth coordinates. Thus, **flight simulation in geographic coordinates should always use double-precision computations and high-order numerical integration algorithms with appropriate time steps.** A good rule of thumb for fourth-order Runge-Kutta flight simulation in geographic coordinates is $0.1 \text{ ft} < V\delta t < 10 \text{ ft}$, where V is the airspeed and δt is the time step. For shorter time steps the round-off error associated with the double-precision computations is excessive and with longer time steps the truncation error for the fourth-order Runge-Kutta algorithm becomes substantial.